Final_Project_Check_In

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install.packages("BiocManager") BiocManager::install("ComplexHeatmap") install.packages("circlize")

Question 1 Data Preparation: Download the Sample RNA-seq Count Matrix and associated Metadata. Ensure that you have both the count matrix (counts.csv) and the metadata file (meta_data.csv) available for your analysis - links below. Gene Selection and Summary Statistics:

```
count <- read.csv("~/Downloads/counts.csv", row.names = 1, check.names = FALSE)
#Bring in the count
data <- read.csv("~/Downloads/meta_data.csv", row.names = 1) #Read in the metatdata

id <- sub("\\..*", "", rownames(count)) #This removes the version number
esr1 <- as.numeric(count[id == "ENSG00000091831", ]) #Counts ESR1
erbb2 <- as.numeric(count[id == "ENSG00000141736", ]) #Counts ERBB2
esr1_log <- log2(esr1 + 1) #Log2 transform
erbb2_log <- log2(erbb2 + 1)

dir.create("figures", showWarnings = FALSE) #Folder for LaTeX
```

Question 2 Gene Selection and Summary Statistics: A) Select One Gene: choose a gene from the dataset that interests you. B) Generate Summary Statistics: Using the count data from the selected gene, compute and report summary statistics, such as mean, median, standard deviation, minimum, and maximum.

```
dpm1 <- as.numeric(count[id == "ENSG00000000419", ])

summary_statistics <- data.frame( #These are the summary statistics for the genes
  Gene = c("DPM1", "ESR1", "ERBB2"),
  Mean = c(mean(dpm1), mean(esr1), mean(erbb2)),
  Median = c(median(dpm1), median(esr1), median(erbb2)),
  SD = c(sd(dpm1), sd(esr1), sd(erbb2)),
  Min = c(min(dpm1), min(esr1), min(erbb2)),
  Max = c(max(dpm1), max(esr1), max(erbb2))
)

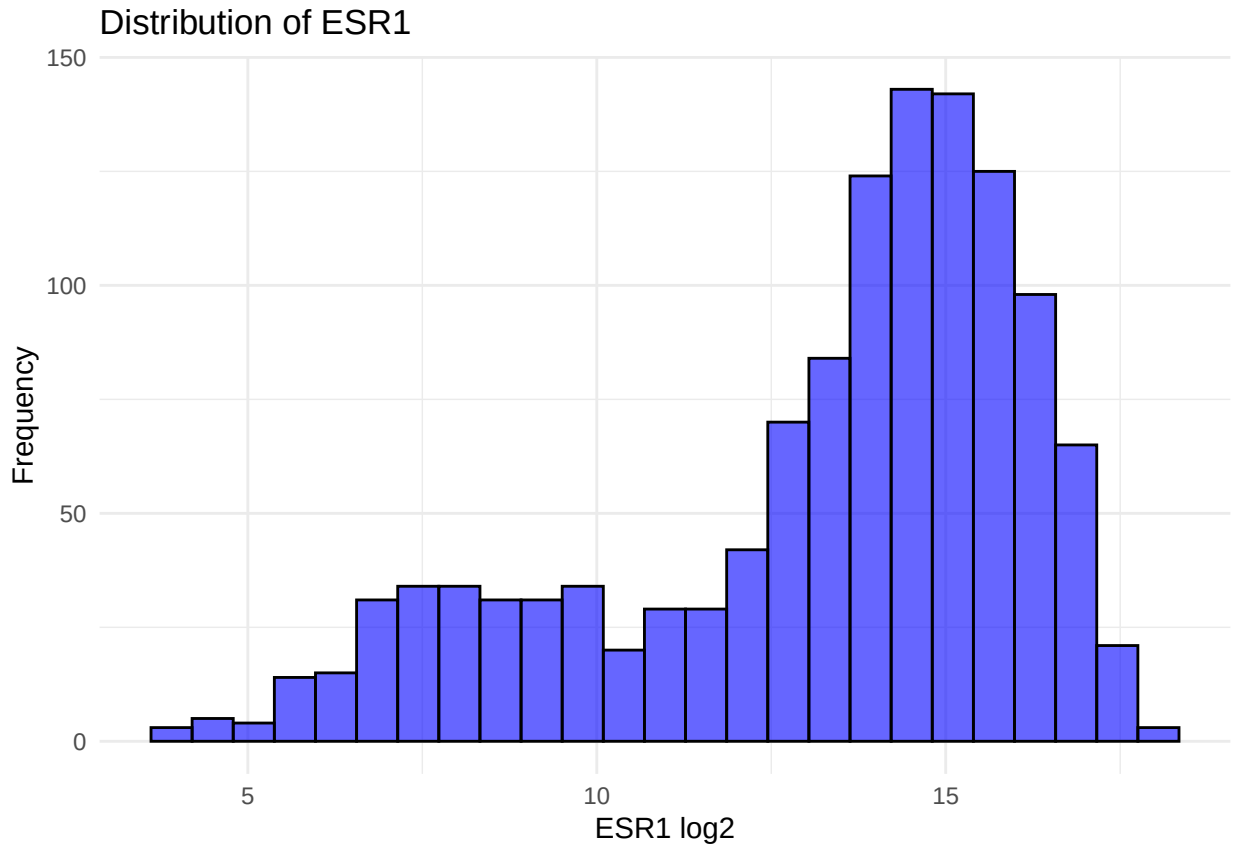
kable(summary_statistics, digits = 1,
      caption = "Sumamry statistics for DPM1, ESR1 & ERBB2")
```

Table 1: Sumamry statistics for DPM1, ESR1 & ERBB2

Gene	Mean	Median	SD	Min	Max
DPM1	2416.3	2052	1459.5	312	17569
ESR1	31442.2	17561	38261.2	14	270977
ERBB2	46982.5	19125	105287.5	205	995067

3.Visualization: A) Create a Histogram: Use ggplot2 to generate a histogram of the count data for the selected gene. This visualization should effectively display the distribution of the counts. B) Create a Scatter Plot: Select a second gene from the dataset. Create a scatter plot using ggplot2 to compare the count data of the two selected genes. C) Create a Violin Plot: Select one covariate from your metadata. Using the count data from the first gene and the selected covariate, generate a violin plot that illustrates the distribution of count data based on the covariate. For example, if you choose "primary_diagnosis", your plot should display a violin plot for each level in "primary_diagnosis". D) Create a Heatmap: Select 10 genes: Choose a set of 10 different genes from the count matrix for your heatmap. Generate a Heatmap: Use the ComplexHeatmap package in R to create a heatmap of the count data for the selected genes. Add an Annotation Bar: Include an annotation bar reflecting your chosen covariate for further context and interpretation of the data.

```
histogram_plot <- ggplot(data.frame(count = esr1_log), aes(x = count)) + #Selected ESR1
  geom_histogram(bins = 25, fill = "blue", #25 counts, blue fill
                 color = "black", alpha = 0.6) + #Black outline
  labs(title = "Distribution of ESR1", #Title is gene name
       x = "ESR1 log2", #X-axis label
       y = "Frequency") + #Y-axis labe
  theme_minimal() #Clean theme
histogram_plot
```

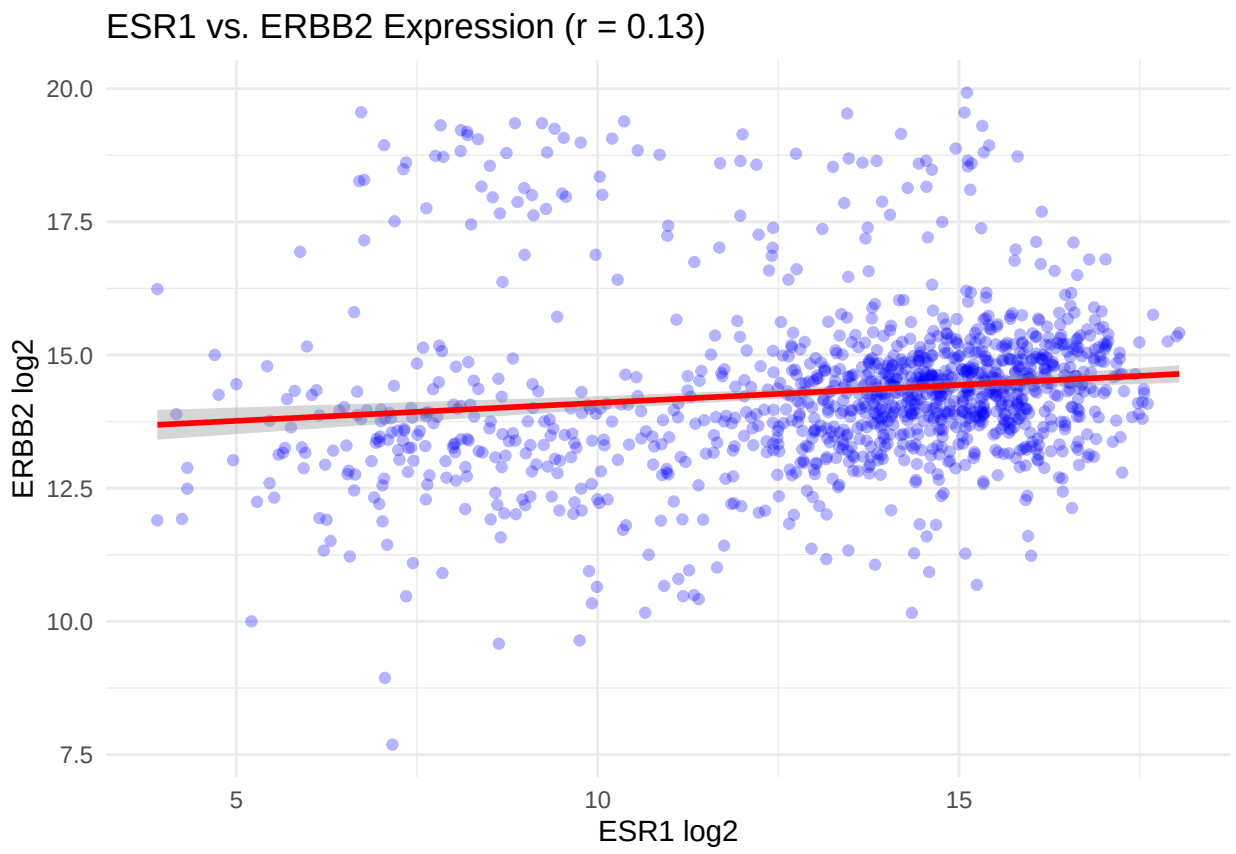



```
ggsave("figures/histogram.png", histogram_plot, width = 6, height = 4, dpi = 300)
#Code for LaTeX report
```

```r
r <- round(cor(esr1_log, erbb2_log), 2)

scatter_plot <- ggplot(data.frame(ESR1 = esr1_log, ERBB2 = erbb2_log), #1 row
                       #per sample
                       aes(x = ESR1, y = ERBB2)) +
  geom_point(alpha = 0.3, color = "blue") +#Blue transparent points
  geom_smooth(method = "lm", formula = y ~ x, color = "red") + #Linear trend line
  labs(title = paste0("ESR1 vs. ERBB2 Expression (r = ", r, ")"), #Title and labels
       x = "ESR1 log2",
       y = "ERBB2 log2") +
  theme_minimal()

scatter_plot
```

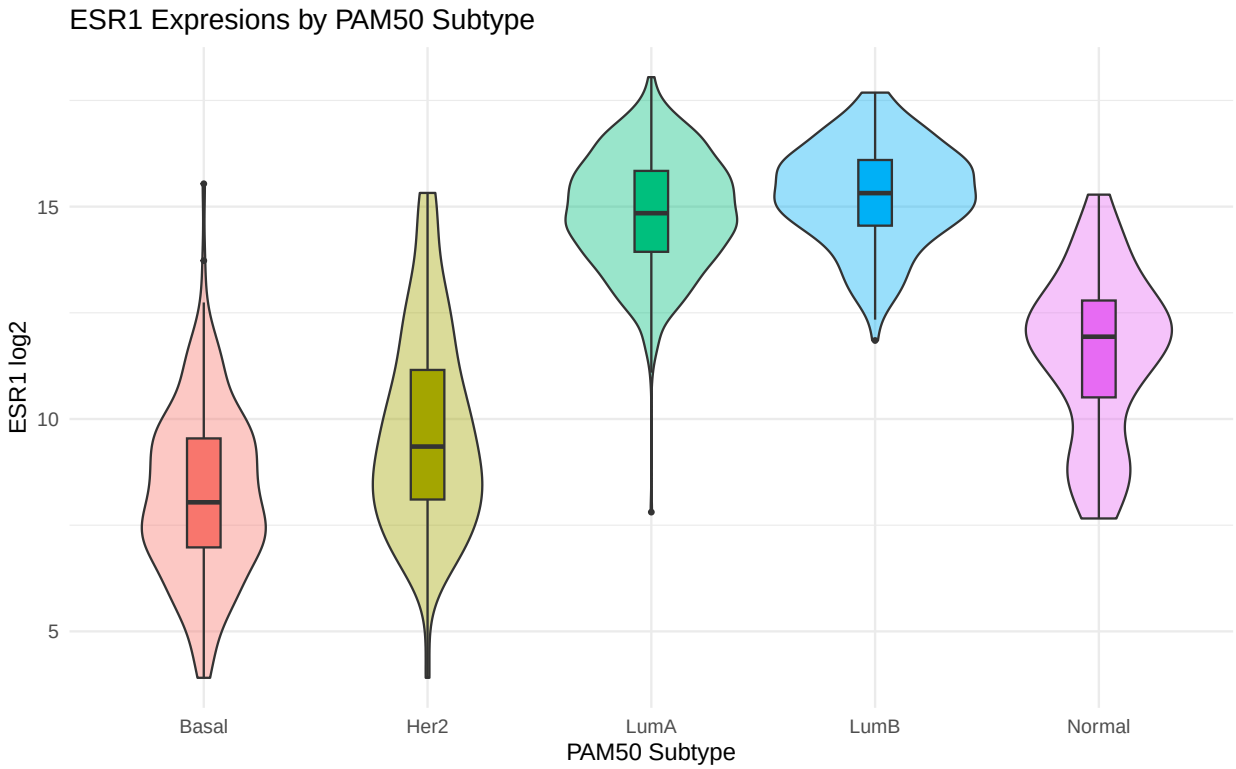



```r
ggsave("figures/scatter_plot.png", scatter_plot, width = 6, height = 5, dpi = 300)
#Save for the LaTeX report
```

```r
box_data <- data.frame( #This is the data for the box plot
  Expression = esr1_log, #ESR1
  Subtype = data$paper_BRCA_Subtype_PAM50) %>% #PAM50 subtype
  filter(!is.na(Subtype)) #Filter out samples without label

box_plot <- ggplot(box_data, aes(x = Subtype, y = Expression, fill = Subtype)) +
  #one box per type
  geom_violin(alpha = 0.4) + #Violin plot
  geom_boxplot(width = 0.15, outlier.size = 0.8) + #Box plot on top
  labs(title = "ESR1 Expresions by PAM50 Subtype", #Title and labels
       x = "PAM50 Subtype",
       y = "ESR1 log2") +
  theme_minimal() +
  theme(legend.position = "none") #removes repeats

box_plot
```




```r
ggsave("figures/box_plot.png", box_plot, width = 6, height = 5, dpi = 300)
#Save file for LaTeX
```

```r
#Utilized https://jokergoo.github.io/ComplexHeatmap-reference/book/ as a source

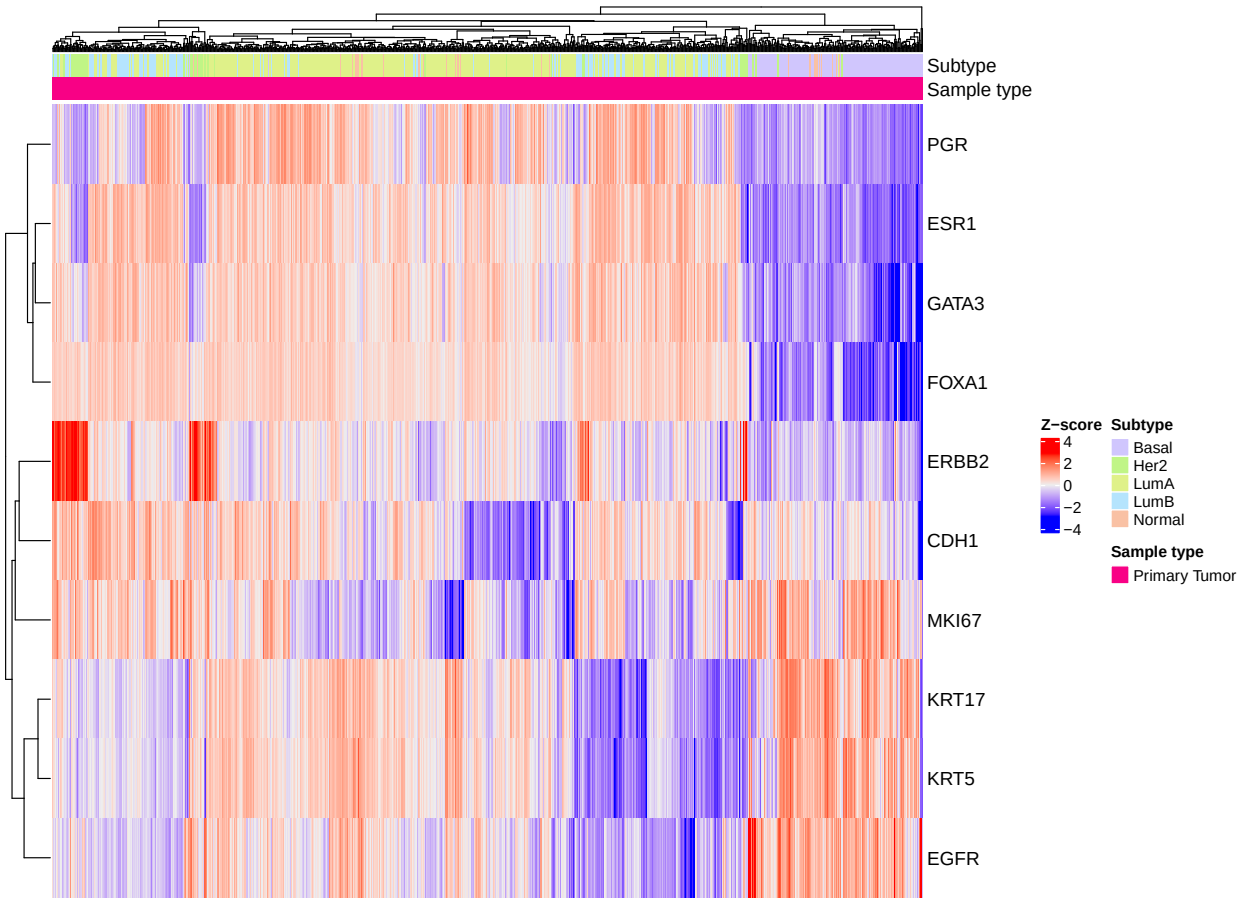
genes <- c(ESR1 = "ENSG00000091831", ERBB2 = "ENSG00000141736", #Picking 10 genes
           GATA3 = "ENSG00000107485", FOXA1 = "ENSG00000129514",
           PGR = "ENSG00000082175", MKI67 = "ENSG00000148773",
           KRT5 = "ENSG00000186081", KRT17 = "ENSG00000128422",
           CDH1 = "ENSG00000039068", EGFR = "ENSG00000146648")
keep <- !is.na(data$paper_BRCA_Subtype_PAM50) #Samples with a label

mat <- log2(as.matrix(count[match(genes, id), keep]) + 1) #Picks the 10 genes,
#log2 transform
rownames(mat) <- names(genes) #Gene symbols as labels
mat <- t(scale(t(mat))) #"Z-score"

column_h <- HeatmapAnnotation( #Annotation bars
  Subtype = data$paper_BRCA_Subtype_PAM50[keep],
  `Sample type` = data$sample_type[keep])

heat <- Heatmap(mat, name = "Z-score", top_annotation = column_h,
                show_column_names = FALSE)

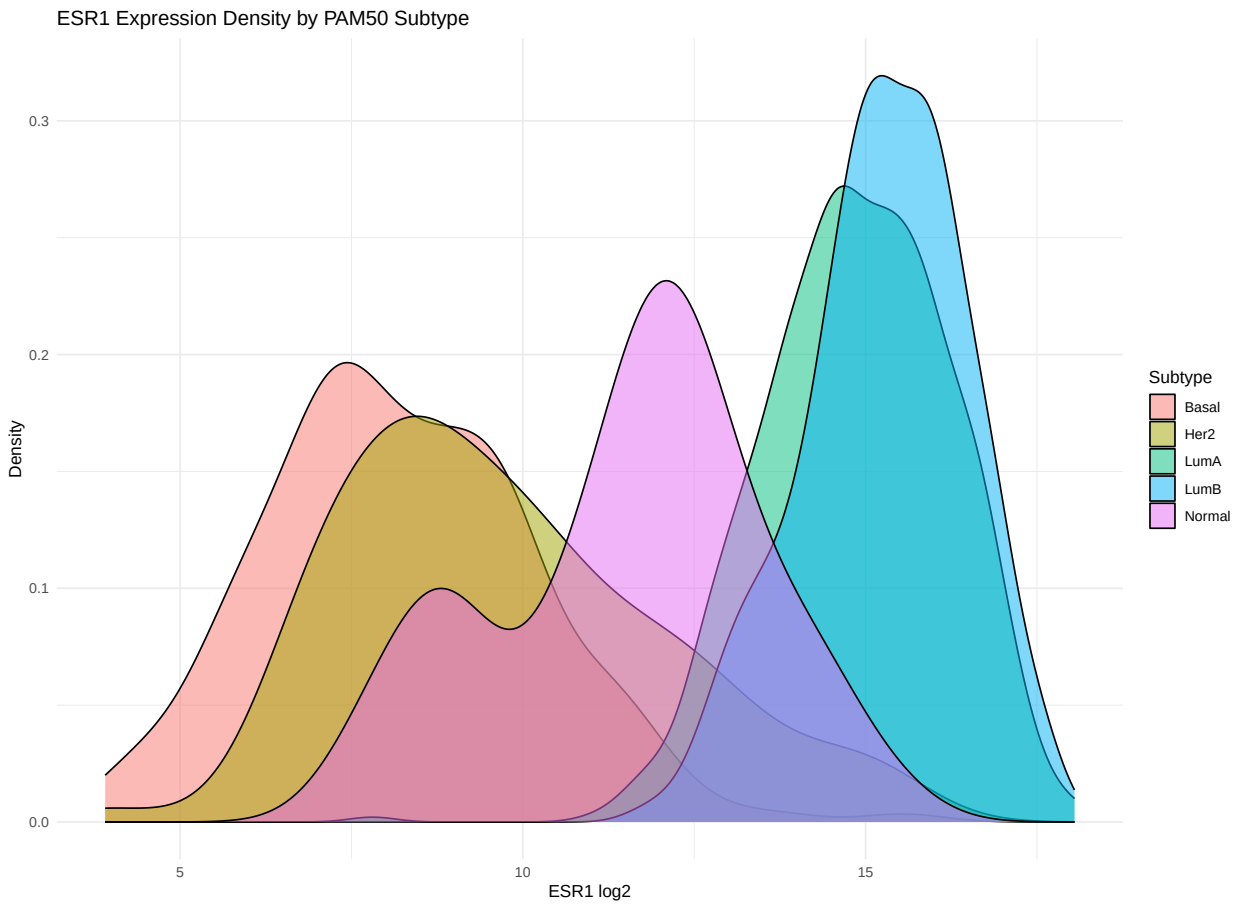
heat
```



```r
png("figures/heatmap.png", width = 2400, height = 1600, res = 300)
draw(heat)
invisible(dev.off())
```

```r
density_plot <- ggplot(box_data, aes(x = Expression, fill = Subtype)) +
  #Continuing the box plot data
  geom_density(alpha = 0.5) + # Smooth transition
  labs(title = "ESR1 Expression Density by PAM50 Subtype", #Title and labels
       x = "ESR1 log2", y = "Density") +
  theme_minimal()

density_plot
```




```r
ggsave("figures/density_plot.png", density_plot, width = 6, height = 5, dpi = 300)
```